

HW 1 Solutions to non-graded problems

1 Problem 2

Bayes' rule says that if $B_1, \dots, B_n$ are a partition of the sample space (i.e. they are disjoint events and their union is the whole space; exactly one of them occurs), then for any set A of positive probability,

$$P(B_k|A) = \frac{P(B_k)P(A|B_k)}{P(A)} = \frac{P(B_k)P(A|B_k)}{\sum_{i=1}^n P(B_i)P(A|B_i)}. \quad (1)$$

If we let H be the event that the selected shoe is hiking, and W the event that it's waterproof, we are trying to find $P(H|W)$. Using (1) gives

$$P(H|W) = \frac{P(W|H)P(H)}{P(Rain)P(W|Rain) + P(H)P(W|H) + P(Rain)P(W|Rain)} \quad (2)$$

$$= \frac{(.7)(.3)}{(.4)(.1) + (.3)(.7) + (.3)(1)} \quad (3)$$

$$= 21/55. \quad (4)$$

2 Problem 3

Let $X_i = 1$ if the i th potato is healthy and 0 otherwise. Then $X = \sum_{i=1}^{10000} X_i$ is the total number of healthy potatoes, and we want to estimate $P(X \geq 8970)$ using the central limit theorem (CLT). The CLT says that if

$$Y = \frac{X - E(X)}{\sqrt{\text{var}(X)}},$$

then Y is approximately distributed like a $N(0, 1)$ random variable if the number of X_i 's is large enough. Thus, we want to translate $X \geq 8970$ into an inequality for Y , and then find the corresponding probability for a $N(0, 1)$ variable using the table in the back of the book. Specifically,

$$\begin{aligned} X \geq 8970 &\iff X - E(X) \geq 8970 - E(X) \\ &\iff \frac{X - E(X)}{\sqrt{\text{var}(X)}} \geq \frac{8970 - E(X)}{\sqrt{\text{var}(X)}} \\ &\iff Y \geq \frac{8970 - E(X)}{\sqrt{\text{var}(X)}}. \end{aligned}$$

The X_i 's are Bernoulli random variables with success probability 0.9, hence with $E(X_i) = 0.9$ and $\text{var}(X_i) = (0.1)(0.9) = 0.09$. Thus $E(X) = \sum_{i=1}^{10000} E(X_i) = 9000$, and since the X_i 's are independent $\text{var}(X) = \sum_{i=1}^{10000} \text{var}(X_i) = 900$. So

$$\frac{8970 - E(X)}{\sqrt{\text{var}(X)}} = -1.$$

Thus, the probability that $X \geq 8970$ should be about $P(Z \geq -1) = P(Z \leq 1)$, where Z is $N(0, 1)$. From Table Va, this is 0.8413.

3 Problem 5

a

The pdf of a $\chi^2(r)$ random variable is

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2},$$

with expectation $\mu = r$ and variance $\sigma^2 = 2r$ (from the inside front cover of textbook). Using $r = 4$, and remembering that $\Gamma(n+1) = n!$ for nonnegative integers n , we get

$$f(x) = \frac{1}{4} x e^{-x/2},$$

$$\mu = 4,$$

$$\sigma^2 = 8.$$

b

We will find the cdf of $Y = X^2$ and then differentiate to find its pdf. Recall that the cdf of Y is $F_Y(t) = P(Y \leq t)$. Since $Y = X^2$,

$$P(Y \leq t) = P(X^2 \leq t) = \begin{cases} 0 & t < 0, \\ P(X^2 \leq t) & t \geq 0. \end{cases}$$

Now for $t \geq 0$, using the fact that $X \sim N(0, 4)$,

$$\begin{aligned} P(X^2 \leq t) &= P(|X| \leq \sqrt{t}) \\ &= P(-\sqrt{t} \leq X \leq \sqrt{t}) \\ &= 2P(0 \leq X \leq \sqrt{t}) \\ &= 2 \int_0^{\sqrt{t}} \frac{1}{\sqrt{8\pi}} e^{-x^2/8} dx. \end{aligned}$$

We now have an expression for the cdf of Y . We can differentiate it wherever it's continuous to get its pdf. For $t < 0$, clearly $F_Y'(t) = 0$, while for $t > 0$, by the fundamental theorem of calculus and the chain rule

$$\frac{d}{dt} 2 \int_0^{\sqrt{t}} \frac{1}{\sqrt{8\pi}} e^{-x^2/8} dx = 2 \cdot (1/2) t^{-1/2} \frac{1}{\sqrt{8\pi}} e^{-(\sqrt{t})^2/8} = \frac{1}{\sqrt{8\pi}} t^{-1/2} e^{-t/8},$$

which is the pdf of $\text{Gamma}(\frac{1}{2}, 8)$ random variable. Thus $Y \sim \Gamma(\frac{1}{2}, 8)$.

c

Using the same calculation as in (b), but for $N(0, 1)$ instead of $N(0, 4)$, you can verify that the square of a $N(0, 1)$ random variable is $\chi^2(1)$. Thus $Z_1^2, \dots, Z_n^2$ are i.i.d. $\chi^2(1)$ random variables, and we want to find the distribution of their sum W . We use moment generating functions. Recall that for any random variable X , its mgf is $M_X(t) = E(e^{tX})$. If $X_1, \dots, X_n$ are independent and $c_1, \dots, c_n$ are constants, then

$$\begin{aligned} M_{c_1 X_1 + \dots + c_n X_n}(t) &= E(e^{t(c_1 X_1 + \dots + c_n X_n)}) \\ &= E(e^{tc_1 X_1} \dots e^{tc_n X_n}) \\ &= E(e^{tc_1 X_1}) \dots E(e^{tc_n X_n}) \\ &= M_{X_1}(c_1 t) \dots M_{X_n}(c_n t). \end{aligned}$$

The mgf of $\chi^2(1)$ is $(1 - 2t)^{-1/2}$ for $t < 1/2$. Since $Z_1^2, \dots, Z_n^2$ are independent, it follows that the mgf of W is

$$M_W(t) = \prod_{k=1}^n M_{Z_k^2}(t) = \prod_{k=1}^n (1 - 2t)^{-1/2} = (1 - 2t)^{-n/2}.$$

This is the mgf of a $\chi^2(n)$ distribution, so $W \sim \chi^2(n)$.

d

Since $Z_i \sim N(\mu_i, \sigma_i^2)$, $(Z_i - \mu_i)/\sigma_i$ is $N(0, 1)$. Thus, this part follows immediately from (c).

4 Problem 6

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - \sum_{i=1}^n \bar{X} = n\bar{X} - n\bar{X} = 0.$$

Next,

$$\begin{aligned}
\sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\
&= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + \sum_{i=1}^n \bar{X}^2 \\
&= \tilde{X} - 2n\bar{X}^2 + n\bar{X}^2 \\
&= \tilde{X} - n\bar{X}^2.
\end{aligned}$$

Therefore,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \tilde{X} - \frac{n}{n-1} \bar{X}^2.$$

5 Problem 7 = 6.1-2 from book

$$\bar{X} = \frac{3.1 + 3.5 + 3.3 + 3.7 + 4.5 + 4.2 + 2.8 + 3.9 + 3.5 + 3.3}{10} = 3.58,$$

$$\begin{aligned}
s^2 &= \frac{1}{10-1} (3.1 - 3.58)^2 + (3.5 - 3.58)^2 + (3.3 - 3.58)^2 + (3.7 - 3.58)^2 + (4.5 - 3.58)^2 \\
&\quad + (4.2 - 3.58)^2 + (2.8 - 3.58)^2 + (3.9 - 3.58)^2 + (3.5 - 3.58)^2 + (3.3 - 3.58)^2 \\
&= 2.356/9,
\end{aligned}$$

$$s = 0.5116.$$

6 Problem 8

The Empirical Rule states that if a dataset is approximately normally distributed (bell-shaped curve), then approximately 68% of the data should lie within one (sample) standard deviation of the (sample) mean, i.e. in the interval $(\bar{x} - s, \bar{x} + s)$, approximately 95% should lie within two standard deviations, i.e. in $(\bar{x} - 2s, \bar{x} + 2s)$, and approximately 99.7% within three standard deviations, i.e. in $(\bar{x} - 3s, \bar{x} + 3s)$. This simply comes from the corresponding values for the $N(0, 1)$ distribution. Specifically, if X is $N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0, 1)$, so

$$P(\mu - \sigma \leq X \leq \mu + \sigma) = P(-1 \leq Z \leq 1) = 2\Phi(1) - 1 \approx 0.6826,$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = P(-2 \leq Z \leq 2) = 2\Phi(2) - 1 \approx 0.9544,$$

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = P(-3 \leq Z \leq 3) = 2\Phi(3) - 1 \approx 0.9974.$$